STAT6611

- Lecture 05 (T, Sept 8)
- Homework 1 assigned
  due F, Sept 18 (F next week)
  by midnight in gradescope/canvas
  (practice on g.i.'s & matrix results)
- My office hours: TW 12:30-1:30 PM
  & R 1-2 PM

TA Ben: R 9:30-10:30 AM

Theorem 2.5. *Under Models II—VII, $\underline{\lambda}'\widehat{\underline{\beta}} = \underline{\lambda}'\widetilde{\underline{\beta}}$ for any two solutions $\widehat{\underline{\beta}}$ and $\widetilde{\underline{\beta}}$ to the normal equations if and only if $\underline{\lambda}'\underline{\beta}$ is estimable.*

Proof. For the "only if" part ($\Longrightarrow$), by Theorem 1.3, all solutions to the normal equations are obtained from

one solution to NEs

$$(X'X)^{-}X'\underline{y} + [I - (X'X)^{-}X'X]\underline{z}$$ ← all other solutions

as $\underline{z}$ ranges over the space of all possible vectors. Thus, setting $\widehat{\underline{\beta}} \equiv (X'X)^{-}X'\underline{y}$ as one solution and $\widetilde{\underline{\beta}} \equiv (X'X)^{-}X'\underline{y} + [I - (X'X)^{-}X'X]\underline{z}$ as another solution for some $\underline{z}$, if $\lambda'\widehat{\underline{\beta}} = \lambda'\widetilde{\underline{\beta}}$ holds for any two solutions of the normal equations, then

$0 = \underline{\lambda}'\widetilde{\underline{\beta}} - \underline{\lambda}'\widehat{\underline{\beta}} = \underline{\lambda}'[I - (X'X)^{-}X'X]\underline{z} = 0$ for all $\underline{z}$.

This implies that

$$\underline{\lambda}'[I - (X'X)^{-}X'X] = \underline{0}' \text{ or } \underline{\lambda}' = \underline{\lambda}'(X'X)^{-}X'X = \underline{a}'X \text{ for } \underline{a} = [\underline{\lambda}'(X'X)^{-}X']'.$$

$\underline{\lambda}' \in R(X)$

By Theorem 2.1, $\underline{\lambda}'\underline{\beta}$ is estimable.

← columns of identity matrix

⇐ If $\underline{\lambda}'\underline{\beta}$ estimable ⇒ $\underline{\lambda}' = \underline{a}'X$ for some $\underline{a}$

So if $\widehat{\underline{\beta}}$ & $\widetilde{\underline{\beta}}$ are two solutions to NEs,

then $X'X\widehat{\underline{\beta}} = X'\underline{y} = X'X\widetilde{\underline{\beta}}$ holds.

or $X\widehat{\underline{\beta}} = X\widetilde{\underline{\beta}}$ by Lemma 1.3 (scratch off)

⇒ $\underline{a}'X\widehat{\underline{\beta}} = \underline{a}'X\widetilde{\underline{\beta}}$ or $\underline{\lambda}'\widehat{\underline{\beta}} = \underline{\lambda}'\widetilde{\underline{\beta}}$

Corollary. $X\widehat{\underline{\beta}} = X\widetilde{\underline{\beta}}$ for any two solutions $\widehat{\underline{\beta}}$ and $\widetilde{\underline{\beta}}$ to the normal equations.

proven above or each row $\underline{u}_i'$ of X is estimable

ASIDE: $\widehat{\underline{\beta}} = (X'X)^{-}X'\underline{y}$ is a solution to NEs. So, any other solution $\widetilde{\underline{\beta}}$, then

so $\underline{y}'\widehat{\underline{\beta}} = \underline{u}_i'\widetilde{\underline{\beta}}$ matches for every row of X

$X\widetilde{\underline{\beta}} = X\widehat{\underline{\beta}} = X(X'X)^{-}X'\underline{y} = P_X\underline{y}$ (Lemma 1.5)

2.4 Geometric Considerations

Definition. Two $t \times 1$ vectors $\underline{a}$ and $\underline{b}$ are said to be orthogonal if their inner product $\underline{a}'\underline{b}$ equals zero.

We adopt the following definitions from Graybill's book *Matrices with Applications in Statistics* (Definition 4.2.4).

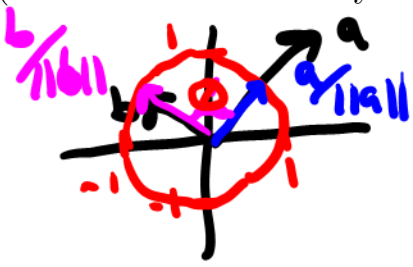
Definition. Let l_1 be the directed line segment from the origin (zero vector) to a non-zero point $\underline{a}$ and let l_2 be the directed line segment from the origin to a non-zero point $\underline{b}$. The angle θ formed by l_1 and l_2 is defined by the formula

$$\cos\theta = \frac{\underline{b}'\underline{a}}{\left[\underline{b}'\underline{b}\right]^{1/2}\left[\underline{a}'\underline{a}\right]^{1/2}}, \quad 0 \le \theta \le \pi.$$

$\left(\frac{\underline{b}}{\|\underline{b}\|}\right)'\left(\frac{\underline{a}}{\|\underline{a}\|}\right)$

$\|\underline{x}\| = (\underline{x}'\underline{x})^{1/2}$

When $\theta = \pi/2$, that is, when $\cos\theta = 0$, that is, when $\underline{a}$ and $\underline{b}$ are orthogonal, the directed line segments (or the lines defined by the directed line segments) are said to be perpendicular.





$\underline{a}'\underline{b} = 0$

"orthogonal complement of C(X)"

Notation. Let $\mathcal{C}^{\perp}(X)$ represent the null space of X', that is, the set of all column vectors that are orthogonal to every column of X [or, equivalently, to every vector in $\mathcal{C}(X)$]. Similarly, $\mathcal{R}^{\perp}(X)$ denotes the set of all row vectors that are orthogonal to every row of X.

$\underline{a} \in \mathcal{C}(X) \Leftrightarrow \underline{X}'\underline{a} = \underline{0}$ where

$X = (\underline{x}_1, \ldots \underline{x}_p)$ $n \times p$

$\underline{X}'\underline{a} = \begin{pmatrix} \underline{x}_1' \\ \vdots \\ \underline{x}_p' \end{pmatrix}\underline{a} = \begin{pmatrix} \underline{x}_1'\underline{a} \\ \vdots \\ \underline{x}_p'\underline{a} \end{pmatrix}$

Remark. Note that

$$\mathcal{C}^{\perp}(X) = \{\underline{a} : \underline{a} \in \mathbb{R}^n, \underline{a}'X = \underline{0}'\} = \{\underline{a} : \underline{a} \in \mathbb{R}^n, X'\underline{a} = \underline{0}\}$$

and

$$\mathcal{R}^{\perp}(X) = \{\underline{b}' : \underline{b} \in \mathbb{R}^p, X\underline{b} = \underline{0}\}.$$

Let $\underline{\widehat{y}} = X\underline{\widehat{\beta}}$ and $\underline{\widehat{\epsilon}} = \underline{y} - \underline{\widehat{y}}$, where $\underline{\widehat{\beta}}$ is any solution to the normal equations.

- elements of $\underline{\widehat{y}}$ are the least squares estimates of corresponding elements of $X\underline{\beta}$.
- elements of $\underline{\widehat{\epsilon}}$ can be regarded as estimates of corresponding elements of $\underline{\epsilon}$.
- the data vector $\underline{y}$ admits the decomposition as the sum of $\underline{\widehat{y}}$ and $\underline{\widehat{\epsilon}}$

$$\underline{y} = \underline{\widehat{y}} + \underline{\widehat{\epsilon}}.$$

- further, the two components $\underline{\widehat{y}}$ and $\underline{\widehat{\epsilon}}$ of this decomposition are orthogonal:

$$\underline{\widehat{y}}'\underline{\widehat{\epsilon}} = (X\underline{\widehat{\beta}})'(\underline{y} - X\underline{\widehat{\beta}}) = \underline{\widehat{\beta}}'X'(\underline{y} - X\underline{\widehat{\beta}}) = \underline{\widehat{\beta}}'(X'\underline{y} - X'X\underline{\widehat{\beta}}) = \underline{\widehat{\beta}}'\underline{0} = 0,$$

The next theorem gives a stronger result regarding the decomposition $\underline{y} = \underline{\widehat{y}} + \underline{\widehat{\epsilon}}$.

Theorem 2.6. *Any $n \times 1$ vector $\underline{y}$ admits a unique decomposition satisfying $\underline{y} = \underline{y}_1 + \underline{y}_2$, where $\underline{y}_1 \in \mathcal{C}(X)$ and $\underline{y}_2 \in \mathcal{C}^{\perp}(X)$. Moreover, this decomposition is given by $\underline{y}_1 = \underline{\widehat{y}} = X\underline{\widehat{\beta}}$ and $\underline{y}_2 = \underline{\widehat{\epsilon}} = \underline{y} - \underline{\widehat{y}} = \underline{y} - X\underline{\widehat{\beta}}$, where $\underline{\widehat{\beta}}$ is any solution to the normal equations.*

Proof. As noted above, $\underline{y} = \underline{\widehat{y}} + \underline{\widehat{\epsilon}}$, and $\underline{\widehat{y}} = X\underline{\widehat{\beta}} \in \mathcal{C}(X)$. Also,

$$X'\underline{\widehat{\epsilon}} = X'(\underline{y} - X\underline{\widehat{\beta}}) = X'\underline{y} - X'X\underline{\widehat{\beta}} = \underline{0}, \text{ so that } \underline{\widehat{\epsilon}} \in \mathcal{C}^{\perp}(X).$$

It remains to verify the uniqueness of the decomposition.

Theorem 2.6 indicates some properties on the geometry of least squares estimation:

1. it holds that $\underline{y}_1 = \widehat{\underline{y}} = X\widehat{\underline{\beta}} = P_X\underline{y}$, where P_X is the projection matrix of X.

   i.e. can compute $\underline{y}_1 = \widehat{\underline{y}} = P_X\underline{y} = X(X'X)^{-1}X'\underline{y}$

   This means that $\widehat{\underline{y}} = X\widehat{\underline{\beta}}$ is the "projection" of the vector $\underline{y}$ onto $\mathcal{C}(X)$; mathematically, it's *the* linear combination of the columns of X that is the closest distance to $\underline{y}$ (i.e., $\|\underline{y} - \underline{t}\|^2 \geq \|\underline{y} - \widehat{\underline{y}}\|^2$ for any other $\underline{t} \in \mathcal{C}(X)$.)

2. as another characterization, if some vector $\underline{t} \in \mathcal{C}(X)$ (i.e., is a linear combination of X columns) and satisfies $X'(\underline{y} - \underline{t}) = 0$ (i.e., the difference $\underline{y} - \underline{t} \in \mathcal{C}^{\perp}(X)$), it must be that $\underline{t} = \widehat{\underline{y}} = X\widehat{\underline{\beta}} = P_X\underline{y}$.

   difference is orthogonal to col. of X

   i.e. $\underline{t} \in \mathcal{C}(X)$ & $X'(\underline{y} - \underline{t}) = 0$

   then $\underline{t} = X\underline{\beta}$ holds with $X'(\underline{y} - \underline{t}) = 0 \Rightarrow X'\underline{t} = X'\underline{y} \Rightarrow X'X\underline{\beta} = X'\underline{y}$

   $\Rightarrow \underline{t} = X\underline{\beta} = \widehat{\underline{y}}$

3. also $\underline{y}_2 = \widehat{\underline{\epsilon}} = \underline{y} - \widehat{\underline{y}} = \underline{y} - X\widehat{\underline{\beta}} = (I - P_X)\underline{y}$, where $I - P_X$ turns out to equal the project matrix (say P_Z) for any matrix Z with columns that span $\mathcal{C}^{\perp}(X)$

   $\underline{\epsilon} = \underline{y} - X\underline{\beta} = \underline{y} - P_X\underline{y} = (I - P_X)\underline{y}$

   This means that $\underline{y} - \widehat{\underline{y}}$ is the "projection" of the vector $\underline{y}$ onto $\mathcal{C}^{\perp}(X)$; it's *the* linear combination of the columns of Z that is the closest distance to $\underline{y}$.

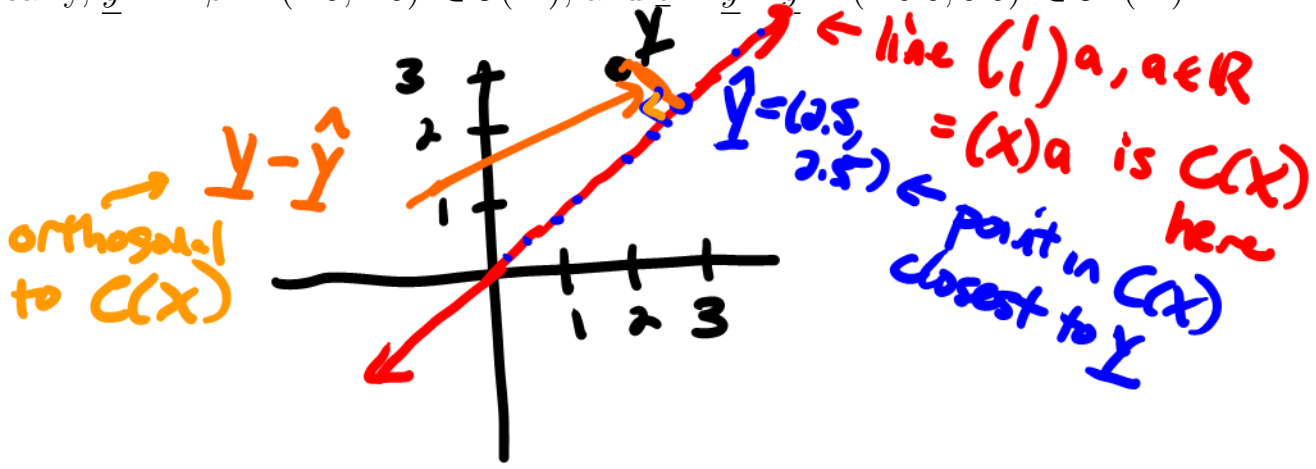
4. in plain words, we can decompose a vector $\underline{y}$ into a "best explanation" $\widehat{\underline{y}}$ of $\underline{y}$ (i.e., geometrically close) as a linear combination of X columns and a "residual" component $\widehat{\underline{\epsilon}} = \underline{y} - \widehat{\underline{y}}$ that cannot be linearly captured by X. These two components are orthogonal/perpendicular.

The following example illustrates there is a unique point $\underline{\widehat{y}}$ in $\mathcal{C}(X)$ such that every line in $\mathcal{C}(X)$ is perpendicular to the line $\underline{y} - \underline{\widehat{y}}$, and this $\underline{\widehat{y}}$ is also the unique point in $\mathcal{C}(X)$ that is closest to (has the minimum Euclidean distance from) $\underline{y}$.

Example. Suppose that

$$n = 2, p = 1, \underline{y} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}, \text{ and } X = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Since $X'X = 2$ and $X'\underline{y} = 5$, the normal "equations" are $2\widehat{\beta} = 5$, which has a unique solution $\widehat{\beta} = 2.5$. For $\underline{y} = (y_1, y_2)'$ plotted on the (y_1, y_2)-plane, the column space of X is the line that passes through the origin and has a slope of 1 in the plane. Clearly, $\underline{\widehat{y}} = X\widehat{\beta} = (2.5, 2.5)' \in \mathcal{C}(X)$, and $\underline{\widehat{\epsilon}} = \underline{y} - \underline{\widehat{y}} = (-0.5, 0.5)' \in \mathcal{C}^{\perp}(X)$.



Corollary to Theorem 2.6. If $\underline{W}$ is any matrix with its column space identical to that of the matrix X [that is, $\mathcal{C}(W) = \mathcal{C}(X)$], then $W\underline{\widehat{\tau}} = X\underline{\widehat{\beta}}$, where $\underline{\widehat{\beta}}$ is any solution to $X'X\underline{\widehat{\beta}} = X'\underline{y}$ and $\underline{\widehat{\tau}}$ is any solution to $W'W\underline{\widehat{\tau}} = W'\underline{y}$.

same least squares estimator

We have $X\underline{\hat{\beta}} = P_X\underline{y}$ for any solution to $X'X\underline{\hat{\beta}} = X'\underline{y}$

& $W\underline{\hat{\tau}} = P_W\underline{y}$ for any solution to $(W'W)\underline{\hat{\tau}} = W'\underline{y}$

Can show $\mathcal{C}(X) = \mathcal{C}(W) \Leftrightarrow P_W = P_X$ (Homework 1)

2.5 The Conjugate Normal Equations

The least squares estimate of an estimable function $\underline{\lambda}'\underline{\beta}$ can be obtained by computing a solution $\underline{\widehat{\beta}}$ to the normal equations and by then forming $\underline{\lambda}'\underline{\widehat{\beta}}$, which is unique and can be written as $\underline{\lambda}'(X'X)^{-}X'\underline{y}$.

$\underline{r} = ((X'X)^{-})'\underline{\lambda}$ or $(X'X)^{-}\underline{\lambda}$ would be a solution to $(X'X)\underline{r} = \underline{\lambda}$ if a solution exists.

Note: We might expect a solution to exist. Why? $\underline{\lambda}'\underline{\beta}$ estimable $\Leftrightarrow \underline{\lambda}' = \underline{g}'X \Leftrightarrow \underline{\lambda} = X'\underline{g}$ $\Rightarrow$ columns of X' & $(X'X)$ are related since $X'XD = X'$ holds for some D

Alternatively, we can compute a solution $\underline{r}$ to the conjugate normal equations

$$X'X\underline{r} = \underline{\lambda}$$

and obtain the least squares estimate of $\underline{\lambda}'\underline{\beta}$ as $\underline{r}'X'\underline{y}$.

Theorem 2.7. *Under Models II—VII, $\underline{\lambda}'\underline{\beta}$ is estimable if and only if there exists an $\underline{r}$ such that $X'X\underline{r} = \underline{\lambda}$; that is, if and only if the conjugate normal equations are consistent.*

Corollary. Under Models II—VII, $\underline{\lambda}'\underline{\beta}$ is estimable if and only if $X'X(X'X)^{-}\underline{\lambda} = \underline{\lambda}$.